15

Positive Definite Matrices

Historical Note

André-Louis Cholesky was a French artillery officer and geodesist who spent his career surveying: triangulation networks in Crete and in North Africa, work that produces enormous least squares problems whose normal equations are symmetric and positive definite. Some time around 1910 he found the factorisation in this chapter and wrote it up in a short manuscript. He never published it. He was killed in action on the last day of August 1918, and the method appeared six years later, in a geodesy bulletin, in a paper by Commandant Benoît explaining what his late colleague had done. It is now run billions of times a day. The other theorem here is older and more famous: Sylvester's law of inertia, from 1852, which says that although a change of coordinates can move the eigenvalues of a quadratic form almost anywhere, it can never change how many are positive and how many negative.

Positive definiteness has been in use since Chapter 13 without ever being given a proper account. It is what makes A^tA invertible, what makes the conjugate gradient method of Chapter 18 work at all, and what Chapter 17 was assuming when it listed Cholesky as the factorisation to use on a symmetric system. This chapter states the characterisations, proves that the factorisation exists, and settles what a change of coordinates can and cannot do to a quadratic form.

15.1 Definitions, and the first test

Definition 15.1 (Definiteness)**.** *Let $A \in M_n(\mathbb{R})$ be symmetric. Call A* positive definite *if $\langle Ax, x\rangle > 0$ for every $x \neq \mathbf{0}$, and* positive semidefinite *if $\langle Ax, x\rangle \geq 0$ for every x. We write $A > 0$ and $A \geq 0$.*

Theorem 14.14 already gave the fundamental test: $A > 0$ exactly when every eigenvalue is positive, and $A \geq 0$ exactly when every eigenvalue is nonnegative. What follows are

the characterisations that are actually usable, since computing eigenvalues to decide definiteness would be absurd.

Proposition 15.2 (Positive definite matrices are exactly the inner products)**.** *If $A > 0$ then $\langle x, y\rangle_A := \langle Ax, y\rangle = x^t Ay$ is an inner product on $\mathbb{R}^n$ in the sense of Definition 12.1. Conversely every inner product on $\mathbb{R}^n$ arises this way from exactly one symmetric positive definite A.*

Proof. Symmetry of the form is symmetry of A; bilinearity is Theorem 4.6; and positive definiteness of the form is Definition 15.1. Conversely, given an inner product $\langle \cdot, \cdot \rangle$, set $a_{ij} = \langle e_i, e_j \rangle$. Bilinearity gives $\langle x, y\rangle = \sum_{i,j} x_i y_j \langle e_i, e_j\rangle = x^t Ay$, symmetry of the form makes A symmetric, and $x^t Ax = \langle x, x\rangle > 0$ for $x \neq \mathbf{0}$ makes it positive definite. Uniqueness holds because a_{ij} is determined as $\langle e_i, e_j\rangle$. □

Key Idea

So the weighted products of Example 12.2(ii) were not a curiosity: they are the diagonal case of the general fact, and choosing an inner product on $\mathbb{R}^n$ is the same act as choosing a positive definite matrix. This is why the quantity $\|e\|_A$ appeared without comment in the analysis of conjugate gradients: it is the norm of the inner product that A itself defines.

15.2 The characterisations

Theorem 15.3 (Equivalent conditions)**.** *For symmetric $A \in M_n(\mathbb{R})$ the following are equivalent.*

(a) A is positive definite.

(b) Every eigenvalue of A is positive.

(c) $A = B^t B$ for some invertible B.

(d) $A = LL^t$ for a lower triangular L with positive diagonal entries, and this L is unique.

Proof. (a) $\iff$ (b) is Theorem 14.14.

(b) $\Rightarrow$ (c). By Theorem 14.7, $A = QDQ^t$ with Q orthogonal and $D = \operatorname{diag}(\lambda_i)$ with every $\lambda_i > 0$. Put $D^{1/2} = \operatorname{diag}(\sqrt{\lambda_i})$ and $B = D^{1/2}Q^t$, which is invertible as a product of invertibles. Then $B^t B = QD^{1/2}D^{1/2}Q^t = QDQ^t = A$.

(c) $\Rightarrow$ (a). $\langle Ax, x\rangle = x^t B^t Bx = \|Bx\|^2$, which is positive for $x \neq \mathbf{0}$ because B is invertible.

(a) $\Rightarrow$ (d). Induct on n, the case $n = 1$ being $A = (a_{11})$ with $a_{11} = \langle Ae_1, e_1\rangle > 0$ and

$L = (\sqrt{a_{11}})$. For $n > 1$ write

$$A = \begin{pmatrix} a_{11} & w^t \\ w & A_1 \end{pmatrix}, \qquad a_{11} > 0,$$

and set $\ell_{11} = \sqrt{a_{11}}$. The *Schur complement* $S = A_1 - ww^t/a_{11}$ is symmetric, and it is positive definite: given $y \neq \mathbf{0}$, take $x = (-w^t y/a_{11},\ y)$, and expanding gives

$$\langle Ax, x\rangle = \frac{(w^t y)^2}{a_{11}} - \frac{2(w^t y)^2}{a_{11}} + y^t A_1 y = y^t\Big(A_1 - \frac{ww^t}{a_{11}}\Big)y = \big\langle Sy, y\big\rangle,$$

and the left side is positive since $x \neq \mathbf{0}$. By induction $S = L_1 L_1^t$, and then

$$L = \begin{pmatrix} \ell_{11} & 0 \\ w/\ell_{11} & L_1 \end{pmatrix} \qquad \text{satisfies} \qquad LL^t = \begin{pmatrix} a_{11} & w^t \\ w & ww^t/a_{11} + S \end{pmatrix} = A.$$

For uniqueness, comparing the $(1,1)$ entries of $LL^t = A$ forces $\ell_{11}^2 = a_{11}$, and positivity of the diagonal picks the positive root; the rest of the first column is then forced by the first column of A, and induction handles L_1.

(d) $\Rightarrow$ (c). Take $B = L^t$, invertible because its diagonal entries are nonzero. □

Figure 15.1 shows what symmetry saves. The two triangular factors of an LU decomposition carry the same information here, so only one need be computed and stored, and that is the whole of the difference between $\frac{1}{3}n^3$ operations and $\frac{2}{3}n^3$.

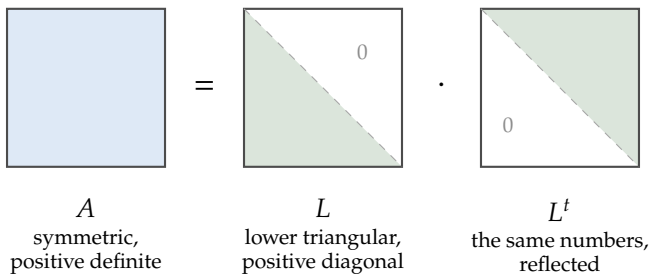


Figure 15.1. The Cholesky factorisation of Theorem 15.3(d). Because A is symmetric the two factors of an LU decomposition carry the same information, so only one need be computed and stored. That is the whole reason Cholesky costs $\frac{1}{3}n^3$ against the $\frac{2}{3}n^3$ of Theorem 8.5, exactly the halving that the cost table in Chapter 17 recorded without explaining.

15.3 Sylvester's criterion, and computing the factor

Condition (d) is a test as well as a factorisation: run the algorithm, and if it never asks for the square root of a non-positive number then the matrix was positive definite. There is also a determinant test.

Lemma 15.4 (Pivots are ratios of leading minors)**.** *Suppose elimination on A can be completed without row swaps, with pivots $d_1, \dots, d_n$. Let A_k be the leading $k \times k$ submatrix. Then* $\det A_k = d_1 d_2 \cdots d_k$ *for every k, so* $d_k = \det A_k / \det A_{k-1}$.

Proof. Elimination subtracts multiples of earlier rows from later ones, and the entries of A_k depend only on rows and columns with index at most k, so the steps that touch A_k are exactly elimination performed on A_k itself. Those are operations of type (R3), which by Proposition 8.3(b) leave the determinant alone, and they leave A_k upper triangular with $d_1, \dots, d_k$ on the diagonal. Now apply Lemma 8.4. □

Theorem 15.5 (Sylvester's criterion)**.** *A symmetric A is positive definite if and only if every leading principal minor is positive:* $\det A_k > 0$ *for* $k = 1, \dots, n$.

Proof. Suppose $A > 0$. For $x \in \mathbb{R}^k$ let $\tilde{x} \in \mathbb{R}^n$ be x followed by zeros; then $\langle A_k x, x \rangle = \langle A\tilde{x}, \tilde{x} \rangle > 0$ for $x \neq \mathbf{0}$, so A_k is positive definite, and by Theorem 15.3(b) and Theorem 9.12 its determinant is a product of positive eigenvalues, hence positive.

Conversely suppose every $\det A_k > 0$. Then no leading minor vanishes, so elimination runs without swaps, and by Lemma 15.4 every pivot $d_k = \det A_k / \det A_{k-1}$ is positive. Elimination without swaps gives $A = LU$ with L unit lower triangular and U upper triangular, by Theorem 4.21; writing $U = DM$ with $D = \operatorname{diag}(d_k)$ the diagonal of U and M unit upper triangular, this reads $A = LDM$. Transposing and using $A^t = A$ gives a second such factorisation, $A = M^t(DL^t)$, with M^t unit lower triangular and DL^t upper triangular. These agree: if $L_1U_1 = L_2U_2$ with both L_i unit lower triangular and both U_i invertible upper triangular, then $L_2^{-1}L_1 = U_2U_1^{-1}$ is at once unit lower triangular and upper triangular, hence $\mathbf{1}$. Here the U_i are invertible because their diagonals are the nonzero d_k. So $M = L^t$ and $A = LDL^t$; putting $B = D^{1/2}L^t$, which is defined because the d_k are positive and invertible because its diagonal is, gives $A = B^tB$, so $A > 0$ by Theorem 15.3(c). □

Caution

The criterion uses the *leading* minors, the ones anchored at the top left, and only those. Testing all principal minors also works but is far more expensive; testing

an arbitrary selection does not work at all. And the criterion is genuinely about positive definiteness: for semidefiniteness it fails, since $\left(\begin{smallmatrix} 0 & 0 \\ 0 & -1 \end{smallmatrix}\right)$ has both leading minors equal to zero and is negative semidefinite. For semidefinite testing one must check every principal minor, not just the leading ones.

Proposition 15.6 (Cost and stability of Cholesky)**.** *Computing L takes about $\frac{1}{3}n^3$ operations. Moreover no pivoting is needed: the entries of L satisfy*

$$\sum_{j \le i} \ell_{ij}^2 = a_{ii}, \qquad \text{so} \qquad |\ell_{ij}| \le \sqrt{a_{ii}}.$$

Proof. The cost is the LU count of Theorem 8.5 halved, since only one triangle is computed. The identity is the (i,i) entry of $LL^t = A$, and every term of the sum is nonnegative, so no single term can exceed the total. □

Key Idea

That bound is the reason Cholesky needs no pivoting, and it is worth comparing with Chapter 17. There, Gaussian elimination was backward stable only when the growth factor stayed small, and Wilkinson's example showed it need not. Here growth is impossible: every entry of L is bounded by the square root of a diagonal entry of A, before the computation starts. Cholesky is unconditionally backward stable, it is half the cost of LU, and it doubles as a test for definiteness. Symmetry with positivity is the most rewarding structure a matrix can have.

Example 15.7 (A factorisation by hand)**.** *Take*

$$A = \begin{pmatrix} 4 & 2 & -2 \\ 2 & 5 & -1 \\ -2 & -1 & 6 \end{pmatrix}.$$

The leading minors are 4, $4\cdot 5 - 4 = 16$ *and* $\det A = 4(30-1) - 2(12-2) - 2(-2+10) = 116 - 20 - 16 = 80$, *all positive, so* $A \succ 0$ *by Theorem 15.5. Following the proof:* $\ell_{11} = 2$, *and the first column of* L *is* $(2, 1, -1)$. *The Schur complement is*

$$S = \begin{pmatrix} 5 & -1 \\ -1 & 6 \end{pmatrix} - \frac{1}{4}\begin{pmatrix} 2 \\ -2 \end{pmatrix}\begin{pmatrix} 2 & -2 \end{pmatrix} = \begin{pmatrix} 4 & 0 \\ 0 & 5 \end{pmatrix},$$

which is diagonal, so $L_1 = \operatorname{diag}(2, \sqrt{5})$ *and*

$$L = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 2 & 0 \\ -1 & 0 & \sqrt{5} \end{pmatrix}.$$

Check the middle entry of LL^t*: row* 2 *of* L *against itself is* $1 + 4 = 5$*, which is* a_{22}*; and row* 3 *against row* 1 *is* -2*, which is* a_{31}*. Note also* $\det A = (\det L)^2 = (2 \cdot 2 \cdot \sqrt{5})^2 = 80$*, as computed above.*

15.4 Conditioning, and why it matters here

Figure 15.2 is worth looking at before the definitions. The level sets of $\langle Ax, x\rangle = c$ are ellipses with semi-axes $\sqrt{c/\lambda_i}$, so the eccentricity of every one of them is $\sqrt{\kappa}$, and the condition number can be seen rather than computed.

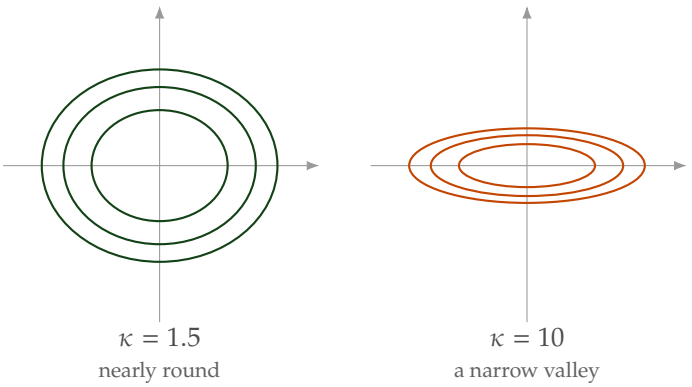


Figure 15.2. Level sets of $\langle Ax, x\rangle = c$ for $c = 1, 2, 3$ and two positive definite matrices. The semi-axes are $\sqrt{c/\lambda_i}$, so the eccentricity of every contour is $\sqrt{\kappa}$. On the left the quadratic of Proposition 18.5 is a round bowl and any descent method walks straight to the bottom; on the right it is a narrow valley and a method that follows the local gradient zigzags across it. The square root is the same one that appears in the conjugate gradient bound of Theorem 18.7, and for the same reason.

Insight

Positive definiteness is what turns solving $Ax = b$ into minimising a bowl, by Proposition 18.5, and the shape of that bowl is governed by κ. Since the contours

have eccentricity $\sqrt{\kappa}$ rather than κ, the natural methods see the square root, which is exactly what Theorem 18.7 reports. It also explains what a preconditioner does geometrically: transforming the problem so that κ falls is transforming the valley into a bowl, and Figure 18.4 is the same statement in terms of the spectrum.

15.5 Sylvester's law of inertia

A change of variables $x = Cy$ turns the form $\langle Ax, x\rangle$ into $\langle C^tACy, y\rangle$. Two symmetric matrices related this way, with C invertible, are called *congruent*. Congruence is not similarity: it does not preserve eigenvalues. It preserves something coarser, and exactly that.

Definition 15.8 (Inertia)**.** *For symmetric A, let n_+, n_- and n_0 be the numbers of positive, negative and zero eigenvalues, counted with multiplicity. The triple (n_+, n_-, n_0) is the* inertia *of A, and $n_+ - n_-$ is its* signature*.*

Lemma 15.9 (A basis-free description of n_+)**.** *n_+ is the largest dimension of a subspace W with $\langle Aw, w\rangle > 0$ for every nonzero $w \in W$.*

Proof. Let V_+, V_-, V_0 be the spans of the eigenvectors for positive, negative and zero eigenvalues, from Theorem 14.7. On V_+, expanding in that eigenbasis gives $\langle Aw, w\rangle = \sum \lambda_i c_i^2 > 0$ for $w \neq \mathbf{0}$, so the maximum is at least $\dim V_+ = n_+$.

Conversely suppose $\dim W > n_+$. Since $\dim(V_- \oplus V_0) = n - n_+$ and $\dim W + (n - n_+) > n$, Exercise 6.31 gives a nonzero $w \in W \cap (V_- \oplus V_0)$. Expanding w in the eigenbasis, only nonpositive eigenvalues occur, so $\langle Aw, w\rangle \leq 0$ and W is not of the stated kind. □

Theorem 15.10 (Sylvester's law of inertia)**.** *Congruent symmetric matrices have the same inertia: if C is invertible then A and C^tAC have the same n_+, n_- and n_0.*

Proof. Write $\tilde{A} = C^tAC$ and note $\langle \tilde{A}y, y\rangle = \langle A(Cy), Cy\rangle$. If W is a subspace on which the form of $\tilde{A}$ is positive, then CW is a subspace of the same dimension, because C is invertible, on which the form of A is positive; and conversely with C^{-1}. So the two maxima of Lemma 15.9 agree and $n_+(\tilde{A}) = n_+(A)$. Applying this to $-A$, whose positive directions are the negative directions of A, gives $n_-(\tilde{A}) = n_-(A)$. Finally $n_0 = n - n_+ - n_-$ in both cases. □

Figure 15.3 shows the difference between congruence and similarity. A congruence may

move the eigenvalues a long way, and cannot move any of them across zero; similarity fixes them exactly. Congruence is the weaker relation, and it is the one appropriate to quadratic forms, where only the signs carry meaning.

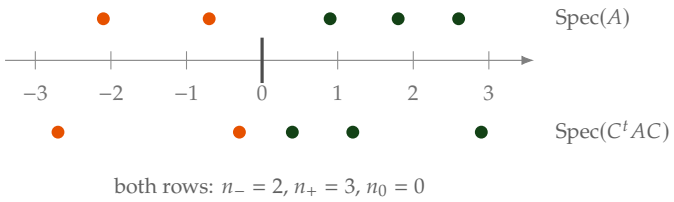


Figure 15.3. Theorem 15.10. A congruence moves the eigenvalues, and it may move them a long way, but it cannot move one across zero. Compare similarity, which fixes the eigenvalues exactly: congruence is the weaker relation appropriate to quadratic forms, where only the sign pattern is meaningful, because the form itself changes with the coordinates while its geometry does not.

Example 15.11 (Diagonalising a form by completing the square)**.** *Take* $q(x) = x_1^2 + 4x_1x_2 + 3x_2^2$*, with matrix* $A = \left(\begin{smallmatrix} 1 & 2 \\ 2 & 3 \end{smallmatrix}\right)$*. Completing the square,*

$$q(x) = (x_1 + 2x_2)^2 - x_2^2,$$

so in the coordinates $y_1 = x_1 + 2x_2$*,* $y_2 = x_2$ *the form is* $y_1^2 - y_2^2$*, with inertia* $(1, 1, 0)$*. Sylvester's law says the eigenvalues of* A *must have the same signs, and indeed* $\det A = -1 < 0$ *forces one of each. Note that neither eigenvalue is* ± 1*: they are* $2 \pm \sqrt{5}$*, roughly* 4.24 *and* -0.24*. The change of variables moved them; it could not change their signs.*

15.6 Where positive definite matrices come from

Proposition 15.12 (Gram matrices)**.** *Let* $v_1, \dots, v_k$ *lie in an inner product space and let* $G_{ij} = \langle v_i, v_j \rangle$*. Then* $G \succeq 0$ *always, and* $G \succ 0$ *exactly when the* v_i *are linearly independent.*

Proof. For $c \in \mathbb{R}^k$, bilinearity gives $\langle Gc, c \rangle = \sum_{i,j} c_i c_j \langle v_i, v_j \rangle = \left\| \sum_i c_i v_i \right\|^2 \geq 0$. It vanishes exactly when $\sum_i c_i v_i = \mathbf{0}$, which for some $c \neq \mathbf{0}$ is precisely dependence. □

Key Idea

That single proposition covers most of the positive definite matrices anyone meets. Taking v_i to be the columns of A gives $G = A^tA$, which is Lemma 13.5 again with the semidefinite case included. Taking them to be centred data vectors gives the covariance matrix of the principal component analysis in Chapter 14, which is why its eigenvalues were guaranteed nonnegative. And in mechanics the stiffness matrix of a structure is positive definite because $\langle Kx, x\rangle$ is the energy stored by the displacement x, and deforming something costs energy.

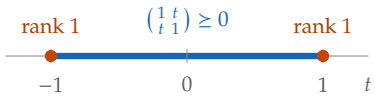
Proposition 15.13 (The semidefinite order)**.** *Write $A \succeq B$ when $A - B \succeq 0$. This is a partial order on symmetric matrices: reflexive, antisymmetric and transitive. It is not total.*

Proof. Reflexivity and transitivity are immediate, since a sum of positive semidefinite matrices is positive semidefinite. For antisymmetry, if $A - B \succeq 0$ and $B - A \succeq 0$ then $\langle (A-B)x, x\rangle = 0$ for every x, and a symmetric matrix with vanishing quadratic form is zero, by Theorem 14.14 applied to its eigenvalues. It is not total: $\left(\begin{smallmatrix}1&0\\0&0\end{smallmatrix}\right)$ and $\left(\begin{smallmatrix}0&0\\0&1\end{smallmatrix}\right)$ are incomparable. □

This order is the setting of semidefinite programming, where one optimises over matrices subject to constraints of the form $A \succeq 0$; it is the natural generalisation of linear programming and the reason positive definiteness is a first-class notion in optimisation rather than only a technical hypothesis.

The sets one optimises over there are called spectrahedra, and the smallest interesting one can be drawn. Figure 15.4 takes the 2×2 symmetric matrices with unit diagonal, where positive semidefiniteness reduces to $1 - t^2 \geq 0$, so the set is a closed segment whose two endpoints are exactly the matrices of rank one.

Open Problem 15.14 (The exact value of Grothendieck's constant)**.** *Let M be a real $m \times n$ matrix. Compare two quantities: the largest value of $\sum_{i,j} M_{ij} s_i t_j$ over signs $s_i, t_j \in \{\pm 1\}$, and the largest value of $\sum_{i,j} M_{ij} \langle u_i, v_j\rangle$ over unit vectors u_i, v_j in a Hilbert space. The second is at least the first, and Grothendieck's inequality says the ratio is bounded by an absolute constant K_G, independent of m, n and M.* What is K_G?
What is known. *It is trapped. Krivine's rounding argument gives $K_G \leq \pi/(2\log(1+\sqrt{2})) \approx 1.7822$, and Braverman, Makarychev, Makarychev and Naor showed that this bound is* not *tight, so K_G is strictly smaller, without saying by how much [107]. From below, Reeds and others give $K_G \geq 1.6769$. So the constant is known to lie in an interval*



a convex set whose extreme points are exactly the matrices that have dropped rank

Figure 15.4. The smallest interesting spectrahedron. A 2×2 symmetric matrix with unit diagonal is positive semidefinite exactly when $\det = 1 - t^2 \geq 0$, that is when $|t| \leq 1$, so the set is a closed segment. Its two endpoints are the matrices of rank one, and they are the extreme points; everything strictly inside has rank two and is a proper average of the two ends. That pattern is general and is the reason semidefinite programming works the way it does: the feasible set is convex, so an optimum may be sought by convex methods, and a linear objective attains its optimum at the boundary, where the rank has dropped. The three-dimensional version, for unit-diagonal 3×3 matrices, is the elliptope, and it is the set Chapter 60 optimises over.

of width about 0.1 *and has been for decades [513].* No closed form is even conjectured with confidence.

Why this is a positive definite question. *The vector version is a semidefinite program: the Gram matrix of the u_i and v_j ranges over the positive semidefinite matrices with unit diagonal, which is the elliptope of this chapter. The sign version is the same problem with the rank constrained to one. So K_G is exactly the worst-case ratio between a semidefinite relaxation and its rank-one truth, and it is the ancestor of every rounding argument in Chapter 60.*

Where a reader might start. *Compute both quantities exactly for $M = \left(\begin{smallmatrix}1 & 1\\ 1 & -1\end{smallmatrix}\right)$. The sign optimum is* 2*; the vector optimum is $2\sqrt{2}$, attained by two pairs of unit vectors at* 45 *degrees. The ratio $\sqrt{2} \approx 1.414$ is a lower bound on K_G from a 2×2 matrix, and improving it is a matter of finding better matrices, which is precisely how the known lower bounds were obtained.*

Exercises

Exercise 15.15. *Decide which are positive definite, using Sylvester's criterion, and check your answers against the eigenvalues.*

(a) $\left(\begin{smallmatrix}2 & -1\\ -1 & 2\end{smallmatrix}\right)$

(b) $\left(\begin{smallmatrix}1 & 2\\ 2 & 1\end{smallmatrix}\right)$

(c) $\left(\begin{smallmatrix}2 & -1 & 0\\ -1 & 2 & -1\\ 0 & -1 & 2\end{smallmatrix}\right)$

Exercise 15.16. *Compute the Cholesky factor of* $\begin{pmatrix} 9 & 3 & -6 \\ 3 & 5 & -4 \\ -6 & -4 & 9 \end{pmatrix}$ *by hand, and verify* $LL^t = A$. *Then obtain* $\det A$ *from* L *without expanding a determinant.*

Exercise 15.17. *Find the inertia of* $q(x) = x_1^2 + 2x_1x_2 + 2x_2x_3$ *by completing squares. Then confirm your answer by computing the signs of the eigenvalues of the associated symmetric matrix.*

Exercise 15.18. *Let* $A > 0$. *Show that* A *has a unique positive definite square root, and that it is a polynomial in* A. Hint: Theorem 14.7 for existence, and Exercise 14.21 for uniqueness; for the last part interpolate the function $\sqrt{\cdot}$ at the eigenvalues.

Exercise 15.19. *Show that if* $A > 0$ *then every diagonal entry is positive, and that* $|a_{ij}| < \sqrt{a_{ii}a_{jj}}$ *for* $i \neq j$. Hint: apply the definition to vectors supported on two coordinates, and recall Theorem 12.3. *Give a symmetric matrix with all diagonal entries positive that is not positive definite.*

Exercise 15.20. *Show that congruence is an equivalence relation, and that* $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 4 \end{smallmatrix}\right)$ *and* $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ *are congruent but not similar. What does that say about which of the two relations quadratic forms care about?*

Exercise 15.21. *For* $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, *verify that* $\langle x, y \rangle_A = x^t A y$ *is an inner product and compute the* A*-norm of* $(1, 0)$ *and the* A*-angle between* $(1, 0)$ *and* $(0, 1)$. *Are they orthogonal in this inner product?*

Exercise 15.22. *Let* $A = \begin{pmatrix} a & w^t \\ w & A_1 \end{pmatrix}$ *with* $a > 0$. *Show that* $A > 0$ *if and only if the Schur complement* $A_1 - ww^t/a$ *is positive definite. This is the step in the proof of Theorem 15.3 isolated as a statement in its own right.*

Exercise 15.23. *Give a symmetric* 2×2 *matrix whose leading principal minors are both nonnegative but which is not positive semidefinite, and explain why this does not contradict Theorem 15.5.*

References and further reading

Cholesky's manuscript and the circumstances of its posthumous publication have been examined in the numerical analysis literature, and the story is worth reading in full; the method is a rare case of a fundamental algorithm whose inventor never knew it had been adopted. Higham [320] gives the sharp backward error analysis for Cholesky, including the precise sense in which the absence of pivoting is safe, and Golub and Van Loan [263] give the blocked variants that are what actually runs. Sylvester's law of inertia is in every account of quadratic forms; Treil [610] is good on it, and on the classification of forms up to congruence that it makes possible. For the semidefinite order and its use in optimisation the literature is large and mostly outside the scope of a linear algebra text; the entry point is any treatment of semidefinite programming, where Proposition 15.13 is the starting definition rather than a closing remark. Banica [51], our baseline, devotes a part of its own to positive matrices, and this chapter is a compressed introduction to it.